

KRISHNA AGARWAL

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PROFESSIONAL SUMMARY

Computer Science & Engineering (Cybersecurity) undergraduate at VIT Vellore with a 9.42/10 CGPA and hands-on experience across Linux, networking, web security fundamentals, software development, databases, and cybersecurity lab environments. Proficient in C++, Python, Java and familiar with security testing workflows, REST APIs, web applications, and backend systems. Building practical experience through cybersecurity labs, ethical hacking training, full-stack applications, and data-intensive software projects.

EDUCATION

Vellore Institute of Technology (VIT), Vellore, India	2025 – 2029
B.Tech in Computer Science & Engineering (Cybersecurity) CGPA: 9.42 / 10.00 VITEEE 2025 AIR: 3,420	
Gujarat Public School, Vadodara, India Higher Secondary Certificate (Class XII) — 83.8%	2025
St. Kabir Indian International School, Vadodara, India Secondary School Certificate (Class X) — 95.6%	2023

RELEVANT COURSEWORK

- Core Systems & Networks:** Operating Systems, Computer Networks, Database Systems, Object-Oriented Programming, Data Structures & Algorithms, Computation Structures
- Applied Computing & Mathematics:** Discrete Mathematics & Linear Algebra, Probability & Statistics, Problem Solving using Python, Problem Solving using Java, Embedded Laboratory Work

TECHNICAL SKILLS

- Cybersecurity:** Linux, Kali Linux, Networking Fundamentals, Web Security Fundamentals, Security Testing Labs, CTF Environments, Nmap, Burp Suite
- Programming:** C++, Python, Java, C, Dart
- Systems & Development Tools:** Linux CLI, Git, GitHub, VirtualBox
- Web & Backend:** React, Next.js, Flask, REST APIs, Firebase, Tailwind CSS, Vercel, Progressive Web Apps
- Databases & Data:** PostgreSQL, pgvector, SQL, Semantic Search, Text Embeddings
- AI/ML:** RAG, Sentence Transformers, Computer Vision, PyMuPDF
- Embedded & IoT:** ESP32, ESP8266, MPU6050, I2C, ThingSpeak, Sensor Interfacing

CERTIFICATIONS & SECURITY TRAINING

EC-Council Hands-on Cyber Security Workshop on CEHv13 — Ongoing

TECHNICAL PROJECTS

Muneem Sahab — Full-Stack Finance Platform & Messaging Bot	Full-Stack Web & Automation
Technologies: React Next.js Flask Firebase REST APIs WhatsApp Business API Tailwind CSS Vercel PWA	
- Built a full-stack finance tracking application with expense categorization, shared/family budget tracking, and responsive web interfaces. - Developed a Flask backend integrated with the WhatsApp Business API and webhook verification for automated bill-splitting workflows. - Implemented Firebase-backed application data handling and deployed the application as a Progressive Web App on Vercel.	
Morrow — Document Intelligence & Retrieval-Augmented Generation System	AI/ML & Database Systems
Technologies: Python PostgreSQL pgvector Sentence Transformers PyMuPDF RAG	
- Built an end-to-end document ingestion pipeline supporting PDF, DOCX, and TXT extraction with overlapping text chunking. - Generated dense text embeddings using Sentence Transformers and implemented semantic similarity retrieval using PostgreSQL and pgvector. - Evaluated retrieval quality using Recall@K and integrated the retrieval layer into a RAG question-answering pipeline.	
Cybersecurity Labs & CTF Practice	Security Testing & Networking
Technologies: Kali Linux Nmap Burp Suite Networking Web Security	
- Practiced Linux-based security testing workflows involving network reconnaissance, service enumeration, and web application analysis in controlled lab and CTF environments. - Used security testing tools such as Nmap and Burp Suite to analyze network services and HTTP application behavior during controlled security exercises.	
Smart Safety Wearable — Real-Time Fall Detection & Indoor Geofencing System	IoT & Embedded Systems
Technologies: Embedded C++ ESP8266/ESP32 MPU6050 I2C Wi-Fi RSSI ThingSpeak	
- Implemented a dual-stage fall-detection algorithm using acceleration magnitude and time-windowed impact detection on an MPU6050 sensor. - Developed Wi-Fi RSSI-based indoor geofencing and ThingSpeak telemetry with a local-first hardware alert architecture.	
AuraGuard — ESP32 Environmental Monitoring System	IoT & Embedded Systems
Technologies: ESP32 Environmental Sensors ThingSpeak C++	
- Developed an ESP32-based environmental monitoring prototype integrating O2, CO, CH4, H2S and PM2.5 sensors with threshold-based alert logic. - Configured ThingSpeak for real-time sensor telemetry, logging and visualization.	
AdaIR — Image Restoration	Computer Vision
Technologies: Python PyTorch Computer Vision	
Worked with the AdaIR image-restoration framework to explore deep-learning-based image enhancement and visual reconstruction.	

ACHIEVEMENTS & ACTIVITIES

- Competitive Programming:** Solved 200+ problems on LeetCode.
- Academic & Entrance Exams:** CGPA: 9.42/10 at VIT Vellore; JEE Main 2025: 96.75 percentile; VITEEE 2025: AIR 3,420.
- Hackathons:** Participant — KLA Hackathon and IQOO Chennai Hackathon.
- GREENWAVE:** Designed and presented an AI-synchronized green-signal corridor concept at a college ideathon.
- Leadership & Service:** Co-organized event scheduling, formats, and technical coordination for World Environment Day Ideathon.